

Isaiah Salcedo

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Seeking an Electrical Engineering internship to apply skills in circuit design, analysis, and problem-solving to contribute to challenging engineering projects.

EDUCATION

Purdue University

Bachelor of Science in Electrical Engineering, GPA: 3.1/4.0

West Lafayette, IN

August 2023 - Present

RELEVANT EXPERIENCE

Latinos in Science and Engineering (MAES), **President**

August 2023 - Present

- Boosted club influence by **leading** community engagement through 12 volunteer events, demonstrating strong communication and teamwork skills.
- Quadrupled club size** from 10 to over 40 members in one semester by creating a welcoming and social environment, resulting in a **300% growth** in student participation.

ASTECH Engineered Products, **ME Intern**

May 2025 - August 2025

- Applied engineering principles to **design** and maintain mechanical components using **Siemens NX/SolidWorks**, improving design efficiency by 15%.
- Authored technical documentation**, including standard work instructions and hardware configurations, to improve production workflows and team communication.

PROJECTS

Purdue Orbital | *Avionics Sub-Team*

August 2024 - Present

- Applied principles of **circuit analysis** and **electronics** to design, simulate (**SPICE**), and lay out custom Printed Circuit Boards (PCBs) with **KiCad** for avionics systems.
- Utilized lab equipment like oscilloscopes for hands-on **troubleshooting** and debugging of electronic circuits, demonstrating strong **problem-solving and analytical skills**.
- Created comprehensive technical documentation, including schematics and interconnect diagrams, to facilitate **teamwork** and cross-functional collaboration.

Purdue Baja Racing | *Suspension Sub-Team*

August 2024 - May 2024

- Assured vehicle safety and effectiveness by directing the sub-team through **CAD modeling** of the suspension system using **SolidWorks**, contributing to a successful, competition-ready vehicle.
- Ensured competition readiness by manufacturing and testing the final suspension assembly, which successfully passed all technical inspections.

Boiler Robotics | *Chassis Robotics Sub-Team*

January 2024 - May 2024

- Engineered a modular prototype system** (tank drive and suspension) utilizing **CAD** to ensure rover stability across diverse terrains, improving performance on uneven ground by 30%.
- Achieved a low center of gravity and efficient maintenance access by **orchestrating multi-team integration**, reducing potential maintenance time by 25%.

Vertically Integrated Projects, NASA Lunabotics Competition | *Tread Lead*

August 2023 - May 2024

- Innovated** a new process for drivetrain tread usage by **leading** a sub-team in the NASA Lunabotics Competition, which aimed to increase traction by 15% on simulated lunar regolith.
- Completed all project deliverables on schedule by **collaborating** with team members on CAD designs and manufacturing tasks, ensuring team synchronization for competition.

TECHNICAL SKILLS

Programming Languages: Python, C, MATLAB

EE Design Tools: LTSpice (SPICE), KiCad (PCB Design), SolidWorks (CAD)

Relevant Coursework: Circuit Analysis, Electronics, Electromagnetics, C Programming, Dynamics